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CAS PCBA 8: Adaptive Cognitive Circuit Engineering with Typed, Fail-Closed Design Rules **1**

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CAS PCBA 8: Adaptive Cognitive Circuit Engineering with Typed, Fail-Closed Design Rules

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Abstract

CAS PCBA 8 is a typed, provenance-bearing research architecture for describing and checking circuit-shaped cognitive-system models. "Circuit" names an engineering organization — typed components, ports, nets, state annotations, calibration references, deterministic traces — and does not assert a physical electrical implementation or identify model quantities with voltage, current, or resistance.

The research question is whether a circuit-shaped modeling language can make cognitive-system hypotheses more explicit, type-safe, replayable, and falsifiable *without* converting an analogy into a physical, causal, safety, or product claim. The candidate supports a qualified affirmative at the software-contract level: semantic identities remain typed; board structure is canonicalized; a deterministic design-rule checker rejects incompatible signal/unit pairs, unresolved ports, unsafe feedback declarations, missing calibrated probes, and unreachable fuse thresholds; supported simulations emit deterministic provenance-rich traces; and only certified finite, deterministic, stateless transitions may cross the TKS v8 adapter boundary.

Two adverse source findings are preserved rather than silently repaired. The published formal CCDL grammar does not cover the book's own five-board reference corpus (four direct syntactic counterexamples). And the published Cognitive Safety Board clamps a red stage at `sat = 0.1` upstream of a fuse threshold of `0.9`, making the stated crossing unreachable under the declared clamp semantics; the unmodified board is retained as a known-fault fixture. At the bound Gate 4 receipt, 82 CAS PCBA v8 test methods pass within an integrated 208-method three-suite run. These receipts establish tested software behavior only — not language completeness, empirical cognitive validity, physical materiality, safety effectiveness, or product readiness.

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1. What "circuit" means here

CAS PCBA treats a candidate cognitive pathway as a directed, typed network whose elements must declare interfaces, allowed values, transfer behavior, statefulness, time semantics, calibration references, and evidence limits. Three separations are load-bearing:

- **Model versus world.** A simulated transfer rule is behavior of the declared software profile, not evidence that a cognitive or physical system obeys it.
- **Design value versus measurement.** A locally supplied parameter or calibration label remains structural metadata until authenticated measurement evidence and uncertainty are supplied.
- **Analogy versus identity.** Borrowed circuit names do not make CAS quantities SI electrical quantities.

The engineering literature supports this discipline at the level of method, not CAS validity: VHDL for machine-readable hardware description [E1], Statecharts for explicit state/hierarchy/concurrency semantics [E2], and the verification-versus-validation distinction for simulation models [E8].

2. Typed semantic and board model

2.1 State types

Building on the CAS Core 8 semantic-identity kernel, the candidate keeps distinct:

```
StateLiteral = Noetic × Domain × optional Drive × optional Register
NDCell       = Noetic × Domain
SimplexAddress40 = strictly positive normalized profile over the 40 NDCells
```

Projection `StateLiteral -> NDCell` forgets Drive and Register and is explicitly many-to-one — not an isomorphism. A simplex address is neither a one-hot cell nor an element of the finite TKS kernel. The mixed-simplex geometry retains a named Fisher-Rao profile; the disclosed binary64 implementation is a numerical pseudometric at its resolution, not a proof that all distinct profiles are separated.

Projector geometry is a separate optional research profile that caps and versions its tolerance, records validation residuals, and binds matrix, basis/source, dimension, rank, and tolerance in one digest.

2.2 Values and arithmetic

Finite carrier values use tagged exact type-plus-value identity, so integer `1`, Boolean `true`, and string `"true"` are distinct throughout DRC, execution, traces, and adapters. All decimal operations run under one immutable, versioned context created from primitive constants; ambient caller precision is irrelevant, and the context identity enters reports and results.

2.3 Board structure

A board contains immutable, identity-bearing records for components and kinds; typed ports (direction, signal kind, unit); typed parameters and finite-state annotations; nets and endpoints; test points and calibration references; declared feedback loops; and source/research-boundary metadata. Board identity includes the complete canonical structure.

3. The design-rule checker

The DRC computes one deterministic diagnostic report under a closed `PCBA8-*` catalog. Errors fail the board; warnings never become an implicit clean result. Checks cover five classes:

1. **Structural integrity** — unique nonblank identities; closed component kinds and exact parameter keys; missing or unknown semantics fail closed.
2. **Typed connectivity** — every endpoint resolves; exactly one typed driver per net; driver/load signal kinds and units agree under a closed admissibility table; required inputs connect or have an executable default in the selected simulator profile.
3. **Calibration and probe integrity** — executable test points require versioned calibration references with profile-digest shape, instrument and protocol identity, uncertainty, measurement kind, and net-unit compatibility; a locally constructed calibration record cannot make itself Evidence Packet eligible; safety-marked paths require marked probes, but a probe's presence does not establish safety.
4. **Feedback analysis** — a directed component graph with strongly connected components (classical algorithms [E3]); each realizable declared cycle needs an exact feedback declaration, declared unit delay, existing dimensionless calibration, and a conservative sub-unity gain bound; unsupported or resource-bounded analysis yields an unresolved-safety diagnostic, never a pass.
5. **Threshold reachability** — conservative upper bounds propagate through causes, bias/gain stages, clipping, and cutoffs; a safety-critical fuse whose threshold exceeds the resolved upstream bound receives `UNREACHABLE_SAFETY_THRESHOLD`; an unresolved bound is not treated as reachable.

4. Adverse source findings

4.1 CCDL grammar coverage

Appendix C's formal `board_block` admits only a small subset of the constructs used by the five published Appendix D / reference boards, which additionally use I/O, CJT, component, rail, dial, `use`, monitor, event, verification, and other forms listed in prose but absent from the EBNF. Pinpoint evidence: the book reserves additional Part IV statement leaders at PDF page 578, calls missing

appendix forms errata at PDF pages 591–592, yet the consolidated EBNF at PDF pages 598–599 still limits `circuit_item` to seven alternatives and a `board_block` body to `layer_decl` or `via_decl`. The machine-readable coverage record documents four direct syntax counterexamples and one bounded listing without an established counterexample; the corpus-level conformance claim therefore fails without overstating the result for any individual board.

Candidate rule: v0.1 declares only its programmatic Python model and does not claim to parse the complete CCDL book language. Any later grammar must publish an exact construct-coverage matrix, reject unsupported forms, and round-trip every admitted board canonically.

4.2 The CSB known-fault fixture

The Cognitive Safety Board listing clamps its red stage at `sat = 0.1` while the downstream fuse threshold is `0.9`; under the stated clamp semantics the fuse cannot trip. The same listing uses non-finite `beta = infinity` despite finite grammar/validator rules, and the book defines no fail-closed latch state machine showing that an open fuse preserves rather than suppresses the refusal path. Evidence: red-stage and fuse values at PDF page 342, repeated at PDF page 608; the Appendix D note at PDF page 610 replaces the infinite gain with "a large finite value" but retains the clamp and does not reconcile it with the threshold.

The unmodified parameters are retained as a known-fault fixture that the DRC must flag. Any repair is a separately versioned design requiring threshold reachability, finite parameters, latch state transitions, fuse truth tables, deterministic traces, hazard analysis, and independent safety review. This is valuable negative evidence about the source and a regression guard for the checker — not a validation of the board's safety function.

4.3 Claim quarantine

The following remain hypotheses or prohibited product inferences: exact thought/transistor isomorphism; shared four-region identity; overload-induced semantic involution; Fisher-Rao causal gain; physical electrical identity for CAS quantities; semantic correctness from a response-region label; safety effectiveness, certification, regulatory or clinical use; complete compiler conformance; production readiness; and disclosure or license authority from a self-hash or generated receipt.

5. Narrow deterministic simulation

Simulation is admitted only for supported component profiles with declared ports, transfer rules, clipping, statefulness, and time semantics. The current profile topologically orders the non-delay graph, uses declared unit-delay behavior to break admitted feedback, and emits a canonical trace binding board and component identities, the exact DRC report, component catalog and decimal-context identities, simulator profile and source baseline, typed inputs/outputs, component events and tick linkage, test-point observations, measurement IDs with parent/child derivations, and the fixed Ring-4 boundary. Measurement construction rechecks child existence, derivation acyclicity, tick linkage, units, measurement kinds, and digest form — a discipline that resembles the entity/activity/derivation separation of W3C PROV [E5] without claiming PROV-O conformance.

The correct reading of a green fixture is: *the software performed the declared typed transition and produced the declared trace for the exact fixture under the exact profile* — never that the fixture proves a cognitive mechanism, causal relation, physical law, safe refusal, or product capability.

6. TKS v8 and Noetic Topos 2 boundaries

Only a certified total, finite, deterministic, stateless, non-stochastic component transition may be adapted to a TKS v8 `FiniteArrow`. Chains become `SemigroupoidPath` values; only an endomap has powers. The adapter certificate binds the exact TKS kernel bytes, board, component, transition table, tagged carrier, semantic profile, and research boundary; ambient module substitution is rejected. The adapter proves at most type-correct representation in the finite reference kernel — it carries no automatic CAS semantic theorem. Continuous simplex profiles, response curves, instrumented observations, stateful devices, stochastic kernels, and memristive history variables remain outside the finite arrow and require distinct types and comparison maps.

From Noetic Topos 2.0 the candidate inherits the ambient-versus-resolved evidence distinction: raw or partially resolved records may be treated as external evidence associated with the ambient construction, but a marked cover does not make a measurement true, and records do not enter resolved semantics until restriction and gluing under one calibration profile are defined and tested.

7. Bounded software evidence

At the bound commit, the integrated Gate 4 receipt records 54 hash-bound artifacts and three green focused suites:

Suite	Test methods	Test-ID SHA-256
CAS Core v8	61	`E2AF5014DE28DD0702211EF32A2757B99B869`
CAS PCBA v8	82	`D5F776C5F15758CD390470B5A542CCA110784`
CAS-Ohm v1	65	`2E81A685961286D944894A22EB93345B97211`

The combined run comprises 208 test methods and 272 parameterized subtests, with 11 claim records all Ring 4 / `NOT LICENSED`, 53 review criteria (20 open), four direct EBNF counterexamples, and eight CAS-Ohm source conflicts. The repository-wide collector was not clean at the bound receipt (four pre-existing package/path namespace collisions after 1,200 collected items), so the focused suites cannot support a repository-wide verification claim.

8. Threats to validity

- **Construct:** no experiment establishes that the selected CAS types correspond to cognitive mechanisms or latent model structure.
- **Internal:** bounded regression suites, not proofs of all paths; property-based and fuzz campaigns [E9], independent implementations, and a separately sourced mutation-testing campaign remain open; the full CCDL parser does not exist.
- **External:** no live calibrated model, physical board, instrument chain, held-out replication, or multi-model evaluation is in the admitted evidence.
- **Numerical:** independent audit of the decimal profile and the binary64 Fisher-Rao pseudometric remains open.
- **Causal:** connectivity, gain composition, prediction, and residuals do not establish causation; causal claims require explicit models, identification assumptions, and interventions [E11]. None is admitted.

- **Safety and governance:** the CSB contradiction blocks any safety-effectiveness inference; hazard analysis, independent safety engineering, regulatory/clinical review, authenticated claim authority, canon admission, disclosure approval, and product acceptance are all open.

9. Dependency-ordered research program

1. Specify a complete versioned CCDL grammar and a golden five-board corpus.
2. Implement canonical parse/serialize round trips; fail closed on unsupported syntax.
3. Define separate capacitor, PLL, memory, power, feedback, stochastic, and memristive state/time profiles.
4. Run a property-based/random-generation campaign over DRC and simulation invariants [E9]; separately specify any mutation-testing campaign.
5. Reproduce the exact source and test bundle in a hermetic toolchain.
6. Preregister calibration and held-out validation through the Gate 5 control kernel without promoting Gate 5 status.
7. Require executor/auditor numerical agreement, falsifiers, negative outcomes, and independent replication.
8. Complete internal cross-review against stable hashes; then commission external formal, systems, experimental, safety, and claim-language review.

10. Conclusion

CAS PCBA 8's value is not that it proves cognition is electrical — it does not and cannot at this stage. Its value is that it turns informal pathway diagrams into typed, inspectable, versioned software objects and *preserves negative evidence* when a source or design cannot satisfy its own contract: a grammar that does not cover its own corpus, and a safety board whose fuse cannot trip, are recorded as first-class findings with page-anchored evidence rather than patched away. The strongest current result is architectural: every analogy is forced through a typed interface, and every executable claim is bound to a profile, source, calibration record, and trace.

References

Identifiers `[E1]`–`[E15]` resolve to complete metadata in `WAVE1_ENGINEERING_CITATION_LEDGER_v0.1.json`.

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